

Production Order List — CSV Export — Detailed Design Document (詳細設計書)

002_DD-CSV — implements [002_BD-CSV](#) version 2, requirements REQ-085–REQ-089.

Document control (改版履歴)

Field	Value
Document ID	002_DD-CSV
Category	UI
System name	ProductionManagementAI
Subsystem name	Production orders
Work item	WI-016
Implements	002_BD-CSV version 2
Created by	Claude (for ThongTM)
Created date	2026-10-07
Last updated by	Claude (for ThongTM)
Last updated date	2026-10-07

Version	Date	Author	Revision content
1	2026-10-07	Claude (for ThongTM)	Initial creation
2	2026-10-07	Claude (for ThongTM)	WI-016 DEC-017: while exporting, item 28 uses <code>aria-disabled="true"</code> with a click guard instead of the <code>disabled</code> attribute, so keyboard focus stays on it (002_BD-CSV E-30; 002_DD-SPD-CSV P-16). Screen item definition, state table and mockup markup updated; nothing else changes

Everything about SCR-002 that this document does not mention stays as [002_DD](#) and its companions specify. Item numbers (28–30), functions (FN-041–FN-043), mappings (M-11–M-18), validations (V-14, V-15) and events (E-30–E-33) are those of 002_BD-CSV and are referenced, not restated.

Overview and reference documents (概要・目次)

Field	Value
File / component name	ExportCsvButton.tsx (new), ProductionOrderListPage.tsx (changed), api.ts , messages.ts , lib/download.ts (new), icons.ts ; backend ProductionOrdersController.Export
Overview	The 「CSV出力」 (Export CSV) control group on SCR-002 and the controller action behind it

Module / method / processing index

No	Name	Overview	Notes
1	ExportCsvButton	Items 28, 29 and 30: button, hint and message, with the export state machine	New component
2	ProductionOrderListPage (change)	Places module 1 in the result header row and resets it when the view changes	Existing route component
3	exportOrdersCsv	Frontend API call: applied view → file, file name and row count	api.ts
4	saveFile	Saves a Blob under a file name through a temporary object URL	New lib/download.ts
5	exportHint	Builds the item 30 text (M-18) from the applied view and total	Pure function in messages.ts
6	ProductionOrdersController.Export	API-PO-05 action: validate, call the service, write the CSV response	Backend; contract in 002_DD-API-CSV

Reference documents

No	Document	Purpose / use	Notes
1	002_BD-CSV	Items, mappings, validations, events, flows	Version 2
2	002_DD and 002_DD-SPD	Existing list modules, view state, live-region conventions	Unedited
3	002_DD-API (API-PO-04)	Parameter names and validation reused by API-PO-05	Unedited
4	002_DB	Indexes the export query relies on	Unedited; no DB change
5	WI-016 decisions.md	DEC-002–DEC-011	

Referenced by

No	Document	Purpose / use	Notes
1	002_DD-API-CSV, 002_DD-FN-CSV, 002_DD-SPD-CSV	Companions (to be written after this file's review)	DEC-011
2	WI-016 test-plan.md	Test viewpoints below	Written in the implementation revision

Component / file organization

X-1. Path structure

No	Path / namespace	Purpose	Notes
1	<code>src/frontend/src/features/production-orders/ExportCsvButton.tsx</code>	Module 1	New
2	<code>src/frontend/src/features/production-orders/api.ts</code>	Module 3	Adds one function
3	<code>src/frontend/src/features/production-orders/messages.ts</code>	MSG-I009, MSG-E024, MSG-E025; <code>labels.list.export*</code> ; module 5	Adds entries only
4	<code>src/frontend/src/lib/download.ts</code>	Module 4	New, feature-independent
5	<code>src/frontend/src/components/icons.ts</code>	<code>Download as ExportIcon</code>	One re-export (WI-004 DEC-023)
6	<code>src/backend/ProductionManagementAI.Api/Controllers/ProductionOrdersController.cs</code>	Module 6	Adds one action
7	<code>src/backend/ProductionManagementAI.Application/ProductionOrders/</code> and <code>... Infrastructure/ProductionOrders/</code>	Export service, CSV writer, repository query	Designed in 002_DD-FN-CSV

X-2. Shared/common components used

No	Name	Purpose	Notes
1	<code>apiClient</code> (<code>src/frontend/src/lib/apiClient.ts</code>)	Same-origin fetch with cookie, 401 → session cleared → <code>/login</code> , <code>ApiError</code> with Problem Details	Module 3 needs a raw <code>Response</code> (blob + headers); 002_DD-SPD-CSV fixes how
2	<code>listViewState</code> (<code>toSearchParams</code>)	The applied view as query parameters	Reused so the export and the list send identical filters
3	<code>iconProps</code>	Decorative 16 px icon	

X-3. Feature-level components used

No	Name	Purpose	Notes
1	ListSummary , PageSizeSelect	Neighbours of item 28 in the result header row	Unchanged

X-4. External APIs used — see "APIs used" below.

X-5. Responsive composition — one component with responsive Tailwind classes. At `sm` and above the button sits in the result header row before 「表示件数」 (Rows per page), with the hint right-aligned under it; below `sm` the button is full width (`w-full` , `min-h-11` = 44 px) on its own row, followed by the hint and the message.

Companion design documents

No	Document	Type	Covers
1	002_DD-API-CSV_製造指示一覽.md	api-design	API-PO-05 <code>GET /api/production-orders/export</code> : parameters, headers, status codes, Problem Details, CSV body and sample file
2	002_DD-FN-CSV_製造指示一覽.md	function-design	Export service method, query reuse without paging, row limit, CSV writer (quoting, BOM, formula neutralising, formats), streaming, logging and tracing
3	002_DD-SPD-CSV_製造指示一覽.md	screen-processing-design	Click → limit check → request → save; error mapping; focus and announcements; reset on view change

Task / design index

No	Category	File-level task	Function/process-level task	Item-level task	Confirmed	Issue	Reviewer	Reworked	Date	Notes
1	basic-design	002_BD-CSV	All	Items 28–30	yes	—	ThongTM	yes (v2, hint)	2026-10-07	DEC-009, DEC-010
2	detailed-design	002_DD-CSV	Modules 1–6, states, mockups	Items 28–30	no	—	ThongTM	no	2026-10-07	This document
3	api-design	002_DD-API-CSV	API-PO-05	—	no	Not yet written	ThongTM	no	—	After review of this file
4	function-design	002_DD-FN-CSV	Export service, CSV writer	—	no	Not yet written	ThongTM	no	—	
5	screen-processing-design	002_DD-SPD-CSV	Export flow	—	no	Not yet written	ThongTM	no	—	

Module design

1. ExportCsvButton

Field	Value
Description	Renders items 28 (button), 30 (hint) and 29 (message) and owns the export state (Ready, Exporting, Succeeded, Failed)
Return type	JSX element
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Preconditions: rendered only when the list query has succeeded with `total` ≥ 1 (BD item 28 display condition). The parent re-mounts it (React `key` = the view key) whenever the applied view changes, which clears item 29 (E-33).

Not applicable — plain component, no rule/validator fields. (The row-limit check V-14 is one comparison inside the click handler, specified in 002_DD-SPD-CSV.)

Arguments

No	Type	Name	Description
1	<code>ListViewState</code>	<code>view</code>	The applied view (from the URL), the same object the list query used
2	<code>number</code>	<code>total</code>	<code>total</code> of the list response that is on screen
3	<code>boolean</code>	<code>disabled</code>	<code>true</code> while a list query is in flight (defensive; the component is normally not mounted then)

Dependencies

No	Module / component	Overview	Notes
1	Module 3 <code>exportOrdersCsv</code>	Performs the request	
2	Module 4 <code>saveFile</code>	Saves the file	
3	Module 5 <code>exportHint</code>	Hint text	
4	<code>message()</code> , <code>labels.list</code>	Catalog text	
5	<code>ExportIcon</code> , <code>iconProps</code>	Download glyph	Decorative

Processing overview: on activation, refuse locally when `total > 10,000`; otherwise call module 3, save the file with module 4 and announce the count, or show the mapped error. The step-by-step flow is in 002_DD-SPD-CSV §1.

Processing flow

Step	Description	Calls
1	Click / Enter / Space on item 28 (E-30)	002_DD-SPD-CSV §1
2	Request and save, or map the error (E-31, E-32)	Module 3, module 4

Return value — None; side effects are the saved file and the item 29 text.

2. ProductionOrderListPage (change)

Field	Value
Description	Adds module 1 to the result header row; nothing else in the page changes
Return type	JSX element
Created by / date	WI-003 / 2026-09-22
Last modified by / date	Claude / 2026-10-07, adds module 1

Preconditions: as 002_DD module 1. Not applicable — plain component, no rule/validator fields.

Arguments — None (route component).

Dependencies — module 1.

Processing overview: inside the existing `page && page.total > 0` branch, the row that holds `ListSummary` and `PageSizeSelect` becomes a wrapping flex row: `ListSummary` left; on the right `ExportCsvButton` then `PageSizeSelect`. `ExportCsvButton` gets `key={viewKey}`, `view`, `total={page.total}` and `disabled={loading}`. Because the branch is not rendered while loading, on the empty state or on the no-match state, item 28 is hidden exactly as BD item 28 requires.

Processing flow

Step	Description	Calls
1	Render module 1 next to <code>ListSummary</code> and <code>PageSizeSelect</code>	Module 1

Return value — None.

3. exportOrdersCsv

Field	Value
Description	<code>exportOrdersCsv(view, signal) → Promise<{ blob: Blob; fileName: string; count: number }></code> — calls API-PO-05 with the view's filter and sort parameters and returns the file
Return type	<code>Promise<ExportedFile></code>
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Preconditions: none. Not applicable — no rule/validator fields.

Arguments

No	Type	Name	Description
1	<code>ListViewState</code>	<code>view</code>	Applied view; <code>page</code> and <code>pageSize</code> are dropped (DEC-002)
2	<code>AbortSignal</code>	<code>signal</code>	Cancels the request when the component unmounts

Dependencies — `listViewState.toSearchParams` (then `page` and `pageSize` deleted), `apiClient` (session handling, `ApiError`).

Processing overview: builds `/api/production-orders/export?{filters&sort}`; on 2xx reads the body as a `Blob`, the file name from `Content-Disposition` (`filename*` first, then `filename`, else a client-built `製造指示一覧_YYYYMMDD-HHmm.csv`) and the row count from the count header defined in 002_DD-API-CSV; on non-2xx throws `ApiError` exactly as other calls do. Header names and status codes: 002_DD-API-CSV.

Processing flow

Step	Description	Calls
1	Build the URL from the view without <code>page</code> / <code>pageSize</code>	<code>toSearchParams</code>
2	Fetch; non-2xx → <code>ApiError</code>	API-PO-05
3	Read blob, file name, count	—

Return value

Type	Name	Description
ExportedFile	—	{ blob, fileName, count }

4. saveFile

Field	Value
Description	saveFile(blob, fileName) : creates an object URL, clicks a temporary <a download> and revokes the URL on the next task
Return type	void
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Preconditions: browser environment. Not applicable — no rule/validator fields. Arguments: Blob blob, string fileName. Dependencies: none. The anchor is never inserted visibly and takes no focus.

5. exportHint

Field	Value
Description	exportHint(view, total) → string : M-18. With hasFilters(view) 「検索した絞り込み条件の{total}件を出力します。」, otherwise 「すべての製造指示{total}件を出力します。」. total is formatted with formatNumber (1,234)
Return type	string
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Rule type: display rule (M-18). Condition reference: hasFilters from listViewState (the same definition that separates the no-match state from the empty state). Check parameters: none.

6. ProductionOrdersController.Export

Field	Value
Description	API-PO-05 action. Same class-level <code>[Authorize(Policy = ProductionOrderEditor)]</code> and <code>[ResponseCache(NoStore = true)]</code> as the list. Validates the query string into the list's typed query, calls the export service and returns its rows as a CSV file result, or a Problem Details error
Return type	<code>Task<IActionResult></code>
Created by / date	Claude / 2026-10-07
Last modified by / date	—

Preconditions: authenticated `Admin / Operator` (policy; else 401/403 by the framework, as for the list). Not applicable — plain controller, no rule/validator fields (validation is `ProductionOrderListQuery.TryCreate`, reused; see 002_DD-FN-CSV).

Arguments: `ProductionOrderExportRequest request` (the list's filter and sort fields, no paging), `CancellationTokens cancellationTokens`.

Dependencies: export service and CSV writer (002_DD-FN-CSV), `ProductionOrderProblems` (Problem Details). Contract: 002_DD-API-CSV.

Screen layout and mockup

DD-level wireframe of the populated state (the BD's SP wireframe is unchanged at this level, so no new SP wireframe):

ProductionManagementAI

製造指示一覧

Admin · ログアウト

製造指示一覧

絞り込み（適用済み：ステータス 進行中・下書き / 納期 2026/10/01～2026/10/31） — 002_DD のとおり

1487件中 1～20件

28↓ CSV出力

15表示件数20

30検索した絞り込み条件の87件を出力します。

29role=status（空） — MSG-I009 / MSG-E024 / MSG-E025 / MSG-E020 の表示位置

指示番号	製品	数量	納期 ▲	ステータス	更新日時
PO-2026-00041	P-1001 — プレーキキャリパー	120 個	2026/10/02	進行中	2026/09/28 09:15
PO-2026-00052	P-2002 — 防錆塗料	12.5 kg	2026/10/03	下書き	2026/09/30 14:02
...					

28: aria-describedby="export-hint" (30). 29 stays mounted, empty, so the first message is announced. Table and paging unchanged.

The six states of the export controls (items 28–30), PC and SP:

A Ready — filters applied

87件中 1～20件

↓ CSV出力

表示件数 20

検索した絞り込み条件の87件を出力します。

B Ready — no filter

87件中 1～20件

↓ CSV出力

表示件数 20

すべての製造指示124件を出力します。

C Exporting — aria-disabled (focus kept), aria-busy

87件中 1～20件

↓ 出力中...

表示件数 20

検索した絞り込み条件の87件を出力します。

D Succeeded — MSG-I009 (role=status)

87件中 1～20件

↓ CSV出力

表示件数 20

検索した絞り込み条件の87件を出力します。

CSVファイルを出力しました（87件）。

E Row limit — MSG-E024 (role=alert)

12,480件中 1～20件

↓ CSV出力

表示件数 20

検索した絞り込み条件の12,480件を出力します。

❗ 出力件数が上限（10,000件）を超えています。条件を絞り込んでから、もう一度お試しください。

F Failed — MSG-E025 (role=alert)

87件中 1～20件

↓ CSV出力

表示件数 20

検索した絞り込み条件の87件を出力します。

❗ CSV出力に失敗しました。もう一度お試しください。

G SP — ready and succeeded (390 px)

87件中 1～20件

表示件数 20

↓ CSV出力

検索した絞り込み条件の87件を出力します。

87件中 1～20件

表示件数 20

↓ CSV出力

検索した絞り込み条件の87件を出力します。

CSVファイルを出力しました（87件）。

Hidden state (list loading, empty, no-match) draws nothing and is not shown. Messages wrap on SP; truncated here only for the sketch.

Mockup: static HTML state gallery in the app's visual vocabulary (Tailwind grey palette of the list page) — [English captions](#) and [Japanese captions](#). Artboards: 1 PC ready with filters, 2 PC ready without filters, 3 PC exporting, 4 PC success, 5 PC row limit, 6 PC failure, 7 SP ready, 8 SP success. Not published as Artifacts (no

publication in plan revision 1).

Region	Contains (field/control)	Notes
Result header row	Item 14 <code>ListSummary</code> ; item 28 <code>ExportCsvButton</code> button; item 15 <code>PageSizeSelect</code>	Wraps; on SP item 28 moves to its own full-width row
Export hint	Item 30	Right-aligned under item 28 on PC; left-aligned under item 28 on SP
Export message	Item 29	Full width under the hint; success in grey, errors in red with the error icon
Table, paging	Items 16–23	Unchanged

Screen item definition

Field	Type	Required	Validation rule	Source (BD ref)
Item 28 「CSV出力」 (Export CSV)	<code><button type="button"></code> with <code>ExportIcon</code> (<code>aria-hidden</code>) + text; classes as the paging buttons (<code>rounded border border-gray-300 bg-white px-3 py-1.5 text-sm text-gray-900</code> , focus ring <code>ring-gray-900</code>), plus <code>inline-flex items-center gap-1.5</code> , <code>max-sm:w-full max-sm:min-h-11 max-sm:justify-center</code> ; <code>aria-describedby="export-hint"</code> ; while exporting <code>aria-disabled="true"</code> + <code>aria-busy="true"</code> with the disabled styling, a click guard and the text 「出力中...」 (Exporting...) — not the <code>disabled</code> attribute, so focus stays on the button (DEC-017); the real <code>disabled</code> attribute only while a list query runs	—	V-14 on click	002_BD-CSV §3 item 28, E-30
Item 30 export hint	<code><p id="export-hint" class="text-xs text-gray-500"></code> , text from module 5	—	—	§3 item 30, M-18
Item 29 export message	Success: <code><p role="status" class="text-sm text-gray-700"></code> MSG-I009. Error: <code><p role="alert" class="flex items-center gap-1.5 text-sm text-red-600"></code> with <code>ErrorIcon</code> + message. Empty <code>role="status"</code> container kept mounted so the first announcement is read	—	—	§3 item 29, E-31, E-32

New catalog entries (`messages.ts`), Japanese text exactly as below:

Key / ID	Japanese	Gloss
<code>labels.list.exportCsv</code>	CSV出力	Export CSV
<code>labels.list.exporting</code>	出力中...	Exporting...
<code>labels.list.exportHintFiltered(total)</code>	検索した絞り込み条件の{total}件を出力します。	Exports the {total} orders matching the searched filters.
<code>labels.list.exportHintAll(total)</code>	すべての製造指示{total}件を出力します。	Exports all {total} production orders.
MSG-I009	CSVファイルを出力しました ({count} 件) 。	Exported {count} orders to CSV.

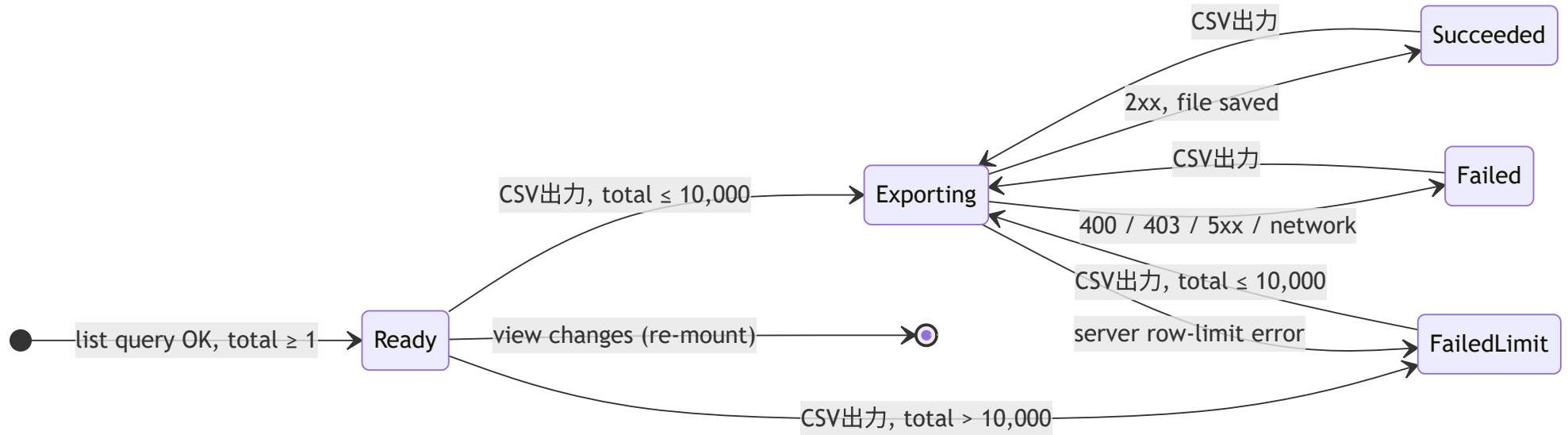
Key / ID	Japanese	Gloss
MSG-E024	出力件数が上限（10,000件）を超えています。条件を絞り込んでから、もう一度お試しください。	The export exceeds the 10,000-order limit. Narrow the filters and try again.
MSG-E025	CSV出力に失敗しました。もう一度お試しください。	The CSV export failed. Try again.

Loading / empty / error / success states

State	Trigger	UI behavior	Data shown
Hidden	List loading, empty (24) or no-match (25)	Items 28–30 not rendered	—
Ready	List shows rows	Item 28 enabled; item 30 shows M-18; item 29 empty	Hint with the shown total
Exporting	Item 28 activated, <code>total</code> ≤ 10,000	Item 28 <code>aria-disabled</code> (activations ignored), <code>aria-busy</code> , text 「出力中…」 ; focus stays on it	Hint unchanged
Succeeded	2xx and the file saved	Item 28 back to 「CSV出力」 ; item 29 <code>role=status</code> MSG-I009 with the response count	「CSVファイルを出力しました（87件）。」
Failed — limit	<code>total</code> > 10,000 (no request), or the server's row-limit error	Item 28 enabled; item 29 <code>role=alert</code> MSG-E024	MSG-E024
Failed — other	403 → MSG-E020; 400 → the list's message IDs from the Problem Details <code>errors</code> ; network, 5xx, aborted body → MSG-E025	Item 28 enabled; item 29 <code>role=alert</code>	Mapped message
Signed out	401	<code>apiClient</code> clears the session; <code>ProtectedRoute</code> sends the user to <code>/login</code> (existing behaviour)	—

Processing and state transitions

State transitions



Transitions from any state on a view change (Search, Clear, sort, page, page size, Back/Forward) and on 401 are table-only.

From state	Event	To state	Side effect
(none)	List query succeeds with <code>total ≥ 1</code>	Ready	Hint rendered (M-18)
Ready / Succeeded / Failed	Item 28 activated, <code>total ≤ 10,000</code>	Exporting	API-PO-05 requested; item 29 cleared
Ready / Succeeded / Failed	Item 28 activated, <code>total > 10,000</code>	Failed (limit)	No request; MSG-E024
Exporting	2xx and body read	Succeeded	File saved (module 4); MSG-I009 announced
Exporting	Row-limit error	Failed (limit)	MSG-E024; nothing saved
Exporting	400 / 403 / 5xx / network / body aborted	Failed	Mapped message; nothing saved
Exporting	401	(signed out)	Session cleared; <code>/login</code>
any	Applied view changes	Hidden, then Ready	Component re-mounted: in-flight export aborted, item 29 cleared, hint rebuilt (E-33)

Processing flows

Flow	Where	Events
Export from the list	002_DD-SPD-CSV §1	E-30, E-31, E-32
Reset on view change	002_DD-SPD-CSV §2	E-33
Produce the CSV	002_DD-FN-CSV	(server side of E-30–E-32)

APIs used

Endpoint	Method	Purpose	Design doc
<code>/api/production-orders/export</code> (API-PO-05)	GET	All orders matching the filters, as CSV	002_DD-API-CSV
<code>/api/production-orders</code> (API-PO-04)	GET	The list itself, unchanged; supplies <code>total</code> for the hint and the limit pre-check	002_DD-API

Database and transaction mapping

Operation	Table(s)	Transaction boundary	Concurrency handling
Read (count + rows)	<code>production_orders</code> , <code>products</code> , <code>production_lines</code>	One read-only transaction for the count and the row query, so the count header and the rows agree	Not applicable — read only; snapshot detail in 002_DD-FN-CSV

No write, no migration, no new index (002_BD-CSV data design overview).

Exception handling

Failure	Retry policy	Timeout	User-facing error	Logging
Over 10,000 rows (client pre-check)	None; the user narrows the filters	—	MSG-E024	None (no request)
Over 10,000 rows (server)	None	—	MSG-E024	Export log entry, outcome <code>limit</code> (002_DD-FN-CSV)
400 invalid parameter	None	—	List message IDs (MSG-E002, MSG-E015–MSG-E019)	As the list

Failure	Retry policy	Timeout	User-facing error	Logging
401 / 403	None	—	/login / MSG-E020	Framework auth logging
Network error, 5xx, connection dropped mid-body	User presses item 28 again	Browser default; no client timeout	MSG-E025	Server: export log entry, outcome <code>failed</code>
User leaves the view during an export	—	—	None (request aborted)	Server sees a cancelled request

Test viewpoints and unresolved decisions

Scenario	Precondition	Expected result	Test-plan ID
Button and hint shown with filters	Filtered list, 87 matches	「CSV出力」 enabled; hint 「検索した絞り込み条件の87件を出力します。」; button described by the hint	TC (implementation revision)
Hint without filters	No filter, 124 orders	「すべての製造指示124件を出力します。」	TC
Hidden on empty / no-match / loading	Each state	No item 28	TC
Export succeeds	≤ 10,000 matches	Request carries filters and sort, no page/pageSize; <code>saveFile</code> called with the header file name; MSG-I009 with the header count	TC
Busy state	Request pending	Button <code>aria-disabled</code> , <code>aria-busy</code> , 「出力中...」, still focused; a second activation sends no request	TC
Client limit	<code>total</code> = 10,001	No request; MSG-E024 as alert	TC
Server errors	403 / 400 / 500 / network	MSG-E020 / list messages / MSG-E025 / MSG-E025; button enabled again	TC
View change resets	Message shown, then Search	Message cleared; hint updated; in-flight request aborted	TC
E2E download PC/SP	Compose stack, demo data	Download event with <code>.csv</code> name; file has BOM, header and the listed rows; axe passes	TC

Unresolved decisions: none for this file. API status codes and header names are decided in 002_DD-API-CSV (BD open questions); the `ApiClient` raw-response approach in 002_DD-SPD-CSV.